

## AI for Research & Science

Three use-cases



Presentation at the NEO-AMELEAS Workshop

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# Outline

## **I) AI for Skeptical People (I)**

Creating Vector Images with TikZ

## **II) AI for Skeptical People (II)**

Finding a Mathematical Counter-Example

## **III) AI for Enthusiasts**

Full Integration (GitHub + Codespaces + Antigravity = Gemini-CLI)

# 01

## AI for Skeptical People (I)

Creating Vector Images with TikZ



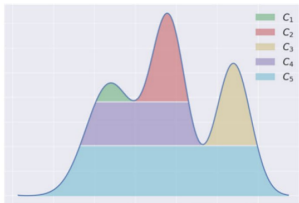
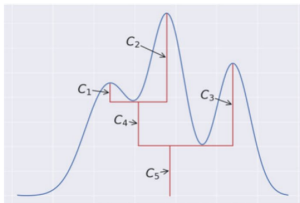


## Enhancing Visual Quality: Fast Vector Graphics with AI

- ▶ **Fast & Professional Figures:** AI allows you to generate high-quality illustrations quickly without wasting time.
- ▶ **What is TikZ?:** A native LaTeX programming language for vector graphics that integrates perfectly with your document.
- ▶ **Full Control:** Since TikZ figures are code-based, you always keep full control to manually edit or fine-tune any detail at any time.



## Example: Target Raster Image

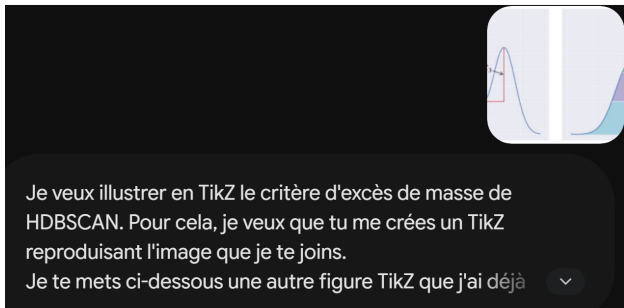


### The Challenge:

- ▶ A 1D density plot illustrating the *excess of mass* in HDBSCAN.
- ▶ Simple raster formats (PNG/JPG) are pixelated and cannot match document fonts.



## Example: Prompting the AI



### Human-AI Collaboration:

- ▶ Simply upload the target image and ask the AI to reproduce it in TikZ.
- ▶ Saves hours of calculating coordinates and writing syntax manually.



## Example: The Generated TikZ Code

### TikZ Code Structure:

```
\begin{tikzpicture}[scale=0.45]
  \tikzset{declare function={f(x) = ...}}

  % Draw density curve
  \draw[thick, blue] plot[domain=0:8] ...;

  % Draw cluster tree bars
  \draw[very thick, red] (1.87,1.70) -- ...;

  % Add labels
  \draw[<-, semithick] (1.87,1.82) -- ...;
\end{tikzpicture}
```

### LaTeX Integration:

#### ► Include Package:

- Add `\usepackage{tikz}` in your preamble.

#### ► Direct Placement:

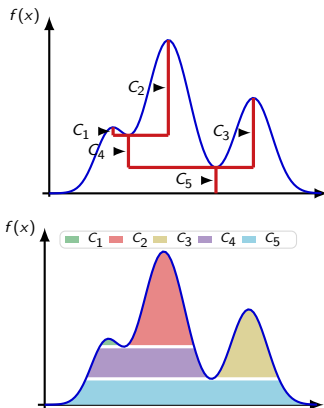
- Paste the generated code directly in the document body.

#### ► No Assets to Manage:

- No need for external image files (PNG/PDF).



## Example: The Native TikZ Result



### Why it works:

- ▶ Native vector rendering means infinite scaling and crispness.
- ▶ Complete control over font size and styling.



# 02

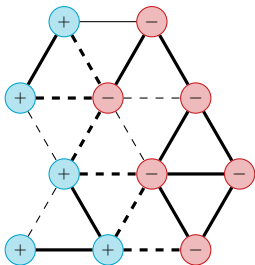
## AI for Skeptical People (II)

Finding a Mathematical Counter-Example





## Case Study: GSBM on Lattices



### Geometric Stochastic Block Model (GSBM):

- ▶ **Ground Truth**  $\Sigma_x \in \{+1, -1\}$ :
  - Node colors represent spins (50% +, 50% -).
- ▶ **Observations**  $Y_e \in \{+1, -1\}$ :
  - Satisfied with probability  $p = 0.75$ .
- ▶ **Graphical Code:**
  - Solid:  $Y_e = +1$  | Dashed:  $Y_e = -1$ .
  - Bold: Satisfied | Thin: Frustrated.
- ▶ **The Goal:** Recover  $\Sigma$  partially (better than random guess)  $\implies$  **Weak Recovery**.



## Case Study: The Long-Path Intuition



### The Correlation Decay Dilemma:

- ▶ For two distant  $+$ -nodes  $x, y$  at distance  $n$ , we infer them as same-community ( $++$ ) only if the path has an **even** number of antiferromagnetic edges.
- ▶ As  $n \rightarrow \infty$ , the probability that the path's sign product matches this true label ( $++$ ) washes out:

$$\mathbb{P}(\text{even}) \xrightarrow{n \rightarrow \infty} \frac{1}{2}$$

- ▶ **The False Conjecture:** Simulations on increasingly larger grids showed the overlap vanishing. This intuition led me to question if weak recovery was truly possible for any noise level  $p_c < 1$ .
- ▶ *The Catch:* The MCMC cluster dynamics simply had an extremely long mixing-time on larger grids, masking the true static phase.



# Case Study: Prompting the AI

 **Manuscrit\_de\_these.pdf**  
PDF

Voici mon manuscrit de thèse en cours de rédaction. Concentre-toi sur le chapitre 11 : détection de communautés sur graphes. Je cherche à montrer (d'abord sur le cas de la grille où  $f_{in} = p_{in} = p = 1 - p_{out}$ ) que le geometric stochastic block model (sous-cas de Sankararaman et Baccelli) n'est pas faiblement recouvrable quel que soit  $p < 1$  arbitrairement proche de 1. L'idée est que  $p_c^{triangle} > p_c^{edges}$ . Et en fait en passant à des higher-order  $m$  encore plus élevés, on montrerait que  $p_c^{(m)} \rightarrow 1$  (il suffirait alors simplement d'évoquer mes théorèmes pour conclure).

Toute la difficulté est de construire des liens/plaquettes d'ordre encore plus élevé puis de montrer que  $p_c^{(m)} \rightarrow 1$ .

Considérons plutôt la grille carré, qui sera peut-être plus simple que la grille triangulaire.

On pave cette grille en damier. Le poids d'une arête est transférée à son unique carré blanc adjacent.

L'idée est de considérer des plaquettes carrées formées de  $(2m+1) \times (2m+1)$  carrés élémentaires. On pave le plan avec ces plaquettes (il y en a deux types).

Il faut alors trouver un moyen de montrer qu'une bonne dynamique sur ces plaquettes fait que lors  $m \rightarrow \infty$ , et en partant de la ground truth, il n'y a pas percolation.

L'idée à  $p$  fixée (température) est de prendre des plaquettes tellement grandes que les états de plus hautes énergie soient tellement plus nombreux qu'il n'y ait pas percolation en une étape de la dynamique. Formalise cela très rigoureusement mathématiquement et montre avec la plus grande rigueur ce résultat.

Afficher moins ^ 

## The Research Question:

- ▶ We uploaded our draft manuscript describing a plaquette dynamics.
- ▶ We asked the AI to formalize the proof that the threshold  $p_c^{(m)} \rightarrow 1$  as the plaquette size  $m \rightarrow \infty$ .
- ▶ If true, this would prove no recovery is possible for any  $p < 1$ .



## Case Study: ChatGPT's Verdict

Réflexion durant 18m 5s >

Il y a un problème assez net avec l'objectif annoncé : **sur la grille carrée, l'énoncé "non faiblement recouvrable pour tout  $p < 1$ " est faux.** Ce n'est pas une petite fissure technique dans le plâtre, c'est plutôt le mur porteur qui vient de déposer sa démission.

### The AI's Answer:

- ▶ The model spent **18 minutes** reasoning.
- ▶ It immediately disproved the theorem: **"on the square lattice, the statement 'no weak recovery for all  $p < 1$ ' is false."**
- ▶ Proving  $p_c^{(m)} \rightarrow 1$  was impossible because the MAP estimator actually achieves recovery.

## Case Study: Why MAP Works (Peierls Contours)

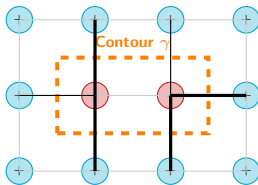
**Peierls Contour Argument:**

- ▶ **Error Contours:** Boundaries separating correct (blue +) and incorrect (red -) spins form loops  $\gamma$  on the dual grid.
- ▶ **MAP Optimality:** The MAP configuration must have a majority of satisfied edges along any contour  $\gamma$ :

$$B(\gamma) \geq \frac{|\gamma|}{2}$$

Otherwise, flipping the spins inside  $\gamma$  would improve the MAP score.

- ▶ **Result:** The probability of having such a contour (implying a majority of noisy edges in the ground truth  $\Sigma$ ) decays exponentially as  $p \rightarrow 1$ , ensuring weak recovery for  $p > p_c$  (where  $p_c$  is the critical threshold).



# 03

## AI for Enthusiasts

Full Integration (GitHub + Codespaces + Antigravity = Gemini-CLI)





# Full Integration (GitHub + Codespaces + Antigravity = Gemini-CLI)





Thank you!